Experiment 4 – Light-emitting diodes

Explanation: goal, principle, procedure and why we need it later

1. The experiment in one sentence

We investigate how an LED behaves electrically (I-V curve), spectrally (colour, line width), energetically (radiant power, efficiency) and spatially (radiation pattern) – and why only a small fraction of the light generated in the chip actually leaves it.

2. Goal of the experiment

- Measure and compare the I-V curves of several LEDs; understand why the forward voltage grows with photon energy (band gap).
- Record emission spectra: determine peak and half-width and compare with theory (≈ 1.8 kT ≈ 47 meV).
- Measure the radiant power versus current (P-I curve) with the integrating sphere; compare the efficiency of "super-bright" and "normal" LEDs.
- Measure the radiation pattern (angular dependence) and compare it with the Lambert radiator.
- Judge white LEDs (RGB, phosphor LED) by their spectra and colour impression.
- Compare theory and measurement and explain deviations with an error analysis.

3. Principle – the logic behind it

3.1 How an LED emits light and what its I-V curve looks like

An LED is a pn junction. Under forward bias electrons (n-side) and holes (p-side) are injected into the active zone and recombine there; the released energy ≈ E_g leaves the crystal as a photon. The curve follows the Shockley equation $I = I_S\,(e^{eU/(nkT)} - 1)$: almost no current up to the "threshold", then an exponential rise, limited at large currents by the series resistance. The threshold is about E_g/e – so an IR LED needs ≈ 1.2 V, a red one ≈ 1.9 V, a blue one ≈ 3 V.

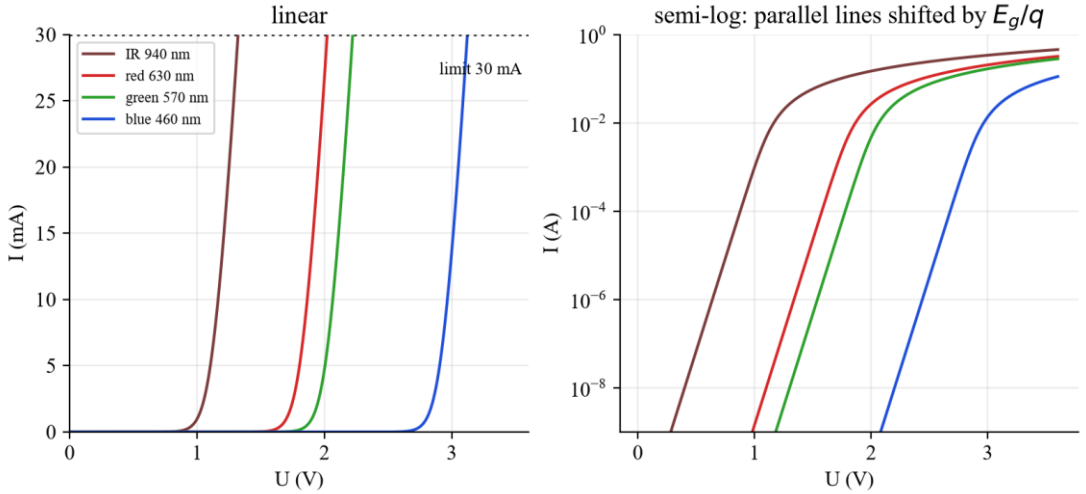


Fig. 1: I-V curves of four model LEDs (LTspice): the curve shifts with photon energy.

3.2 Structure: double heterostructure

Practically all LEDs are double heterostructures: a small-gap material is embedded between two large-gap barriers. Carriers are confined to a layer thinner than 1 µm (instead of the diffusion length > 10 µm in GaAs), so recombination becomes far more probable. The barriers are also transparent to the emitted light, which is therefore not re-absorbed. Quantum wells as active region are even more efficient.

3.3 Spectrum and colour

The ideal spectrum is $I(E) \propto \sqrt{(E - E_g)} \cdot \exp[-(E - E_g)/kT]$: a square-root edge at E_g followed by the thermal (Boltzmann) tail. The maximum is at $E_g + kT/2$, the half-width ≈ 1.8 kT ≈ 47 meV. Conversion to wavelength: λ[nm] = 1240 / E[eV]. A red LED (630 nm) therefore has a theoretical width of only ≈ 15 nm, a blue one (460 nm) ≈ 8 nm; real LEDs are broader (quantum wells, inhomogeneity, heating).

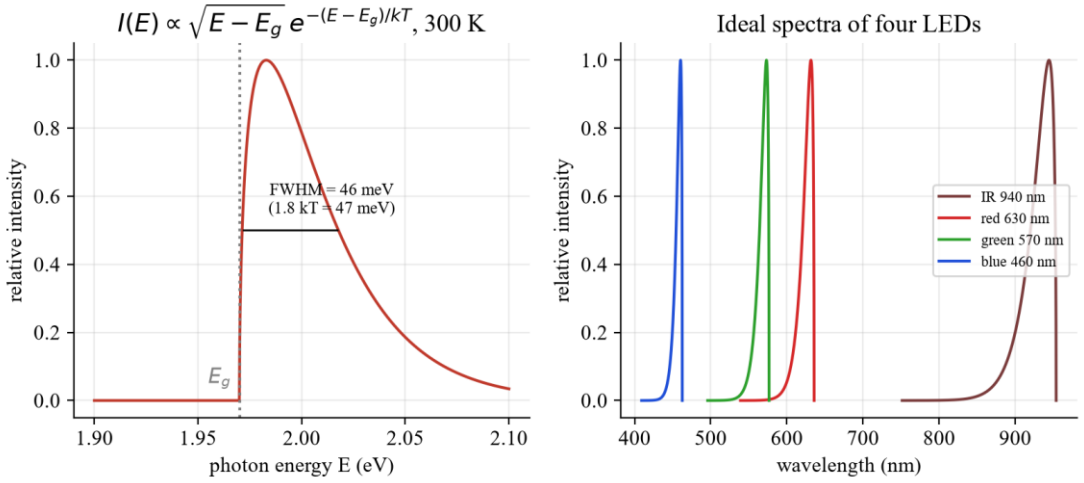


Fig. 2: Idealised spectrum (left) and four LEDs on the wavelength axis (right).

3.4 Efficiencies

- η_{int} = generated photons / injected electrons (quality of the active zone).
- η_{ext} = photons emitted into free space / injected electrons.
- $\eta_{eff} = \eta_{ext}/\eta_{int}$ = outcoupling efficiency – the real bottleneck.
- From the measurement: $\eta_{ext} = P \cdot \lambda[nm] / (1240 \cdot I[A])$; wall-plug efficiency $\eta_{WP} = P / (U \cdot I)$.

3.5 Why does so little light come out?

Semiconductors have high refractive indices (GaAs: $n_1 = 3.5$). At the interface to air ($n_2 = 1$) light is lost twice: (a) **Fresnel reflection** $R = ((n_1 - n_2)/(n_1 + n_2))^2 = 31$ % at normal incidence; (b) **total internal reflection** for incidence angles $\alpha > \alpha_C = \arcsin(n_2/n_1) = 16.6°$. Only a narrow cone escapes: $\eta_{eff} \approx \frac{1}{2}(1 - \cos\alpha_C)(1 - R) \approx \frac{1}{4}(n_2/n_1)^2 (1 - R) \approx$ **1.4 %**. The rest is absorbed in the chip and becomes heat. The plastic **epoxy dome** (n ≈ 1.5) helps twice: a smaller index step ($\alpha_C = 25°$) and a hemisphere with the chip in its centre, where all rays meet the surface perpendicularly – η_{eff} rises to ≈ 4 % (about a factor of 3). Further ideas: thin-film LEDs, textured surfaces, TIP geometry, mirrors.

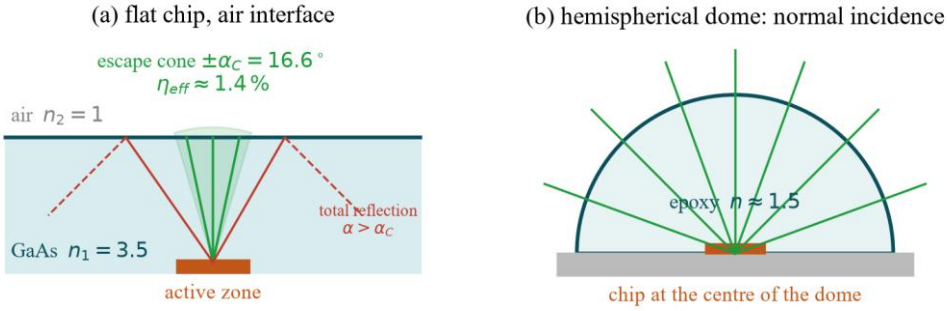


Fig. 3: (a) bare chip: only the ±16.6° cone escapes; (b) hemispherical dome: normal incidence.

3.6 Radiation pattern

A flat diffuse surface radiates according to **Lambert's law** $I(\theta) \propto \cos\theta$ (projection of the area); the half-intensity angle is 60°. An epoxy dome or lens narrows the pattern (e.g. 30° or 15°), described by $\cos^m\theta$. The integrating sphere provides an almost perfect Lambert field at its exit and is therefore used to measure total power independent of the pattern.

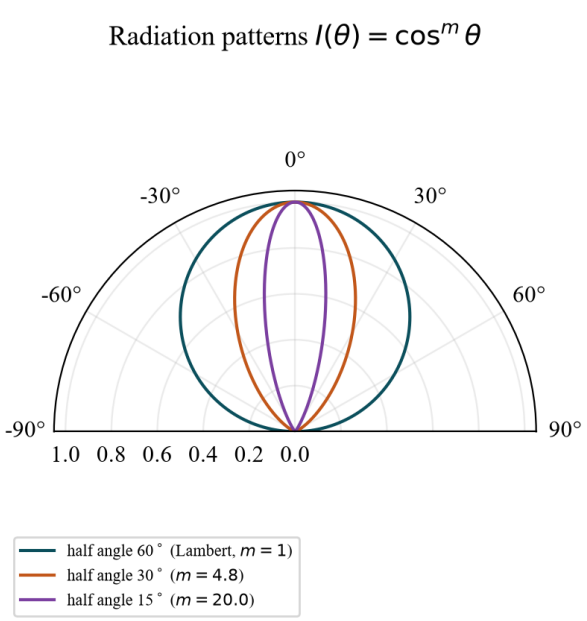


Fig. 4: $\cos^m\theta$ patterns with 60° (Lambert), 30° and 15° half angle.

3.7 Luminous flux and white LEDs

The **luminous flux** $\Phi_{vis} = 683\ lm/W \cdot \int V(\lambda)\, \Phi_e(\lambda)\, d\lambda$ weights the radiant power with the eye sensitivity V(λ): 1 W at 555 nm = 683 lm, but only ≈ 181 lm at 630 nm and ≈ 41 lm at 460 nm. **White light** is made (i) by mixing several LEDs (e.g. R+G+B: narrow lines with gaps → poor colour rendering) or (ii) by a blue LED with phosphor (YAG:Ce, broad yellow-green-red band → a more continuous spectrum, better rendering, but not tunable).

4. Procedure step by step

Equipment: current/voltage source, multimeter, spectrometer with optical fibre (PC), integrating sphere, rotatable holder. The supervisor selects the diodes. **Never exceed 30 mA!**

Task	What to do?	Evaluation
a) I-V	Current 1 µA … 30 mA (logarithmic at low current), U at the multimeter	Semi-log plot, threshold, slope (n), R_S
b) Spectrum	Spectrometer + fibre in front of the LED, subtract dark signal	λ_{peak}, Δλ, $\Delta E = 1240 \cdot \Delta\lambda/\lambda^2$
c) P-I	LED in the integrating sphere, same current steps, signal at the exit	η_{ext}, η_{WP}, bright vs. normal
d) Angle	Chip on the rotation axis, θ = −90…+90° (5°), detector fixed	Half-intensity angle, compare with cos θ
e) White	Spectra of white LEDs, colour impression with test objects	Gaps in the spectrum ↔ colour rendering

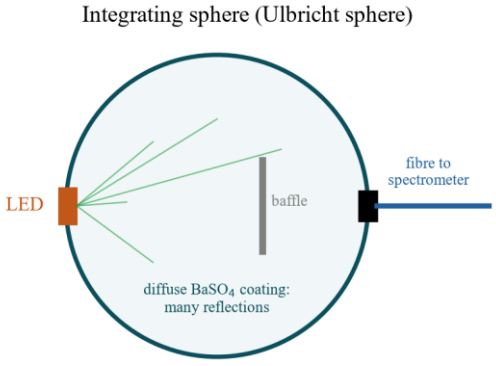


Fig. 5: Integrating sphere with baffle and fibre to the spectrometer.

For preparation there is also a small LTspice demo (optional): the I-V curves of four LEDs and the effect of a series resistor. Files: LED_IV_Kennlinien.asc and LED_Vorwiderstand_Arbeitspunkt.asc (or *04_LTspice/LTspice_starten.bat*).

5. What to watch in the lab

- **Set the current limit ≤ 30 mA** before connecting the LED; mind the polarity (long leg = anode).
- Heating: wait briefly at each point; U falls by ≈ −2 mV/K and the spectrum shifts to longer wavelengths.
- Spectrometer: subtract the dark signal, avoid saturation (adjust integration time), shield ambient light.
- Angle scan: **emitting chip on the rotation axis**, not the package edge; detector in the far field.
- Protocol: for each task "expectation from theory – measured result – explanation of deviations (error analysis)".

6. Why do we need this later?

- **Experiment 5 – laser diodes:** same methods (I-V, P-I, spectrum, beam pattern); a laser diode is an LED with a resonator – its P-I curve additionally has a **threshold** (spontaneous → stimulated emission).
- **Experiment 6 – DFB laser:** the broad LED spectrum (47 meV) is the counterpart to the narrow laser line; mode selection turns it into a single sharp peak.
- **Experiment 1 – free-space transmission:** the LED is the transmitter there; modulation bandwidth and beam angle set the range.
- **Experiment 2 – energy conversion:** solar cell and LED are "the same device in the opposite direction"; the band gap governs both.
- Practice: lighting, displays, indicator lamps – efficiency (lm/W), colour rendering and beam angle are the key figures.

7. Typical questions – short answers

Question	Short answer
Why does an LED light up only in forward direction?	Only there are electrons and holes injected into the active zone and recombine.
Why $U \approx E_g/e$?	The photon has $\approx E_g$; each electron must bring at least that energy.
Why a double heterostructure?	Carriers are confined (more recombination), barriers are transparent.
Why only ≈ 1–2 % outcoupling?	Total reflection outside ±16.6° and 31 % Fresnel loss: $\eta = \frac{1}{4}(n_2/n_1)^2(1 - R)$.
How does the epoxy dome help?	Smaller index step + hemisphere (normal incidence): about a factor 3.
What is a Lambert radiator?	$I \propto \cos\theta$, half-intensity angle 60°.
What is the integrating sphere for?	Measures the total power independent of the radiation pattern.
How do you mount the LED for the angle scan?	Emitting chip exactly on the rotation axis.
Lumen vs. watt?	Lumen = watt weighted with V(λ); 683 lm/W at 555 nm.
RGB white vs. phosphor white?	RGB: narrow lines with gaps → poor rendering; phosphor: broad band → better.
Why max. 30 mA?	Overheating destroys the junction; use a current limit.
How to get the half-width?	$\Delta\lambda$ at half the maximum; $\Delta E = 1240 \cdot \Delta\lambda/\lambda^2$; theory ≈ 1.8 kT ≈ 47 meV.

Date: **18 Nov 2026**, pre-test (Antestat) 14:00, building BC, room BC 119 (group 9).